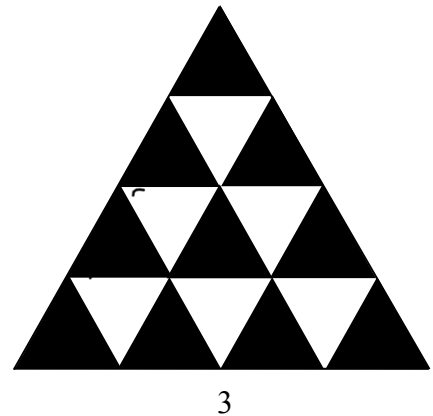
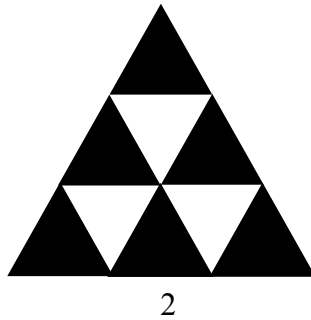
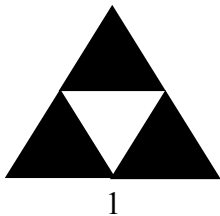
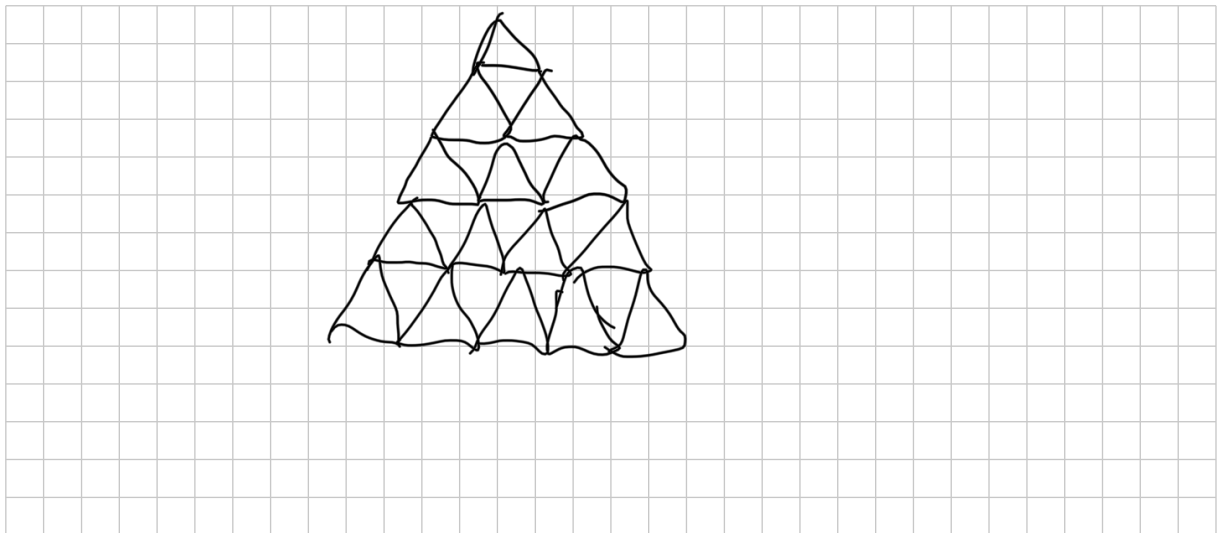


Question 7**(70 marks)**

The first three patterns in a sequence of patterns are shown below.



(a) Draw the fourth pattern in the sequence.

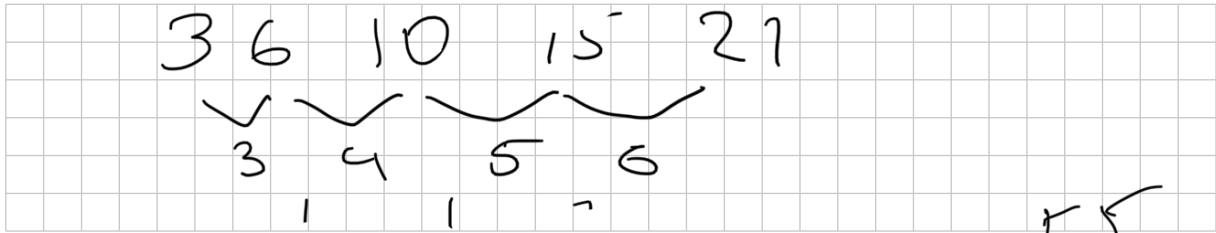


(b) Complete the table below.

	Number of Black Triangles	Number of White Triangles	Total Number of Small Triangles
Pattern 1	3	1	$T_1 = 4$
Pattern 2	6	3	$T_2 = 9$
Pattern 3	10	6	$T_3 = 16$
Pattern 4	15	10	$T_4 = 25$
Pattern 5	21	15	$T_5 = 36$

28
36

- (c) Show that the numbers of **black** triangles form a quadratic sequence.



- (d) (i) How many **black** triangles are in the 9th pattern?

Answer:

55

- (ii) How many **white** triangles are in the 9th pattern?

Answer:

45

- (iii) How many small triangles, in total, are in the 9th pattern?

Answer:

100

- (e) Write an expression in n for the total number of triangles in the n^{th} pattern.

Answer: $T_n = (n+1)^2$

- (f) The number of **black** triangles in the n^{th} pattern is given by the formula $B_n = \frac{1}{2}n^2 + \frac{3}{2}n + c$.

Find the value of c .

$$\frac{1}{2}(1)^2 + \frac{3}{2}(1) + c = 3$$

$$0.5 + 1.5 + c = 3 \Rightarrow -c = -1 \Rightarrow c = 1$$

- (g) Use your answers to parts (e) and (f) above to find a formula for the number of **white** triangles in the n^{th} pattern.

$$W_n = T_n - B_n \Rightarrow (n+1)^2 - \left(\frac{1}{2}n^2 + \frac{3}{2}n + c\right)$$

$$\Rightarrow (n+1)(n+1) - 0.5n^2 + 1.5n + 1$$

$$n^2 + n + n + 1 - 0.5n^2 - 1.5n - 1$$

$$W_n = 0.5n^2 + 0.5n$$

- (h) One particular pattern has a total of 625 triangles. Find the number of black triangles and the number of white triangles in that pattern.

$$T_n = 625 \Rightarrow (n+1)^2 = 625$$

$$\Rightarrow n^2 + 2n + 1 = 625$$

$$B_{24} = \frac{1}{2}(24)^2 + \frac{3}{2}(24) + 1$$

$$\Rightarrow 325 \text{ black } \Delta \Rightarrow n = 24$$

$$W_{24} = 0.5(24)^2 + 0.5(24)$$

$$300 \text{ white } \Delta$$

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